

Math 54 Discussion Worksheet

Name: _____ ID: _____

Problem 1

In the vectorspace of all real functions, find a basis for the space spanned by $\{\sin(t), \sin(2t), \sin(t) \cos(t)\}$.
Hint: Recall the double-angle formula.

Problem 2

Suppose that $T : V \rightarrow W$ is a one-to-one transformation so that $T(v) = T(u)$ implies that $v = u$. Show that if $\{v_1, \dots, v_n\}$ are linearly independent, then so are the vectors $\{T(v_1), \dots, T(v_n)\}$.

Problem 3

Let $\mathcal{E}$ be the standard basis for $\mathbb{R}^n$ and $\mathcal{B}$ be some other basis for $\mathbb{R}^n$. Let $P_{\mathcal{B}}$ be the matrix whose columns are the vectors $b \in \mathcal{B}$. Given a vector v whose coordinates are given in the standard basis, how does one use $P_{\mathcal{B}}$ to find its coordinates in the $\mathcal{B}$ basis? What if instead we were given x with its coordinates in the $\mathcal{B}$ basis? Answer both of these by writing down an appropriate matrix equation.